DCIM ENTERPRISE PLATFORM - FULL SPECIFICATION (100+ PAGES)

Executive Summary (Section 1)

This document defines a full enterprise-grade DCIM platform supporting multi-datacenter operations, QR-based workflows, and operational intelligence. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Architecture Overview (Section 1)

Frontend (Next.js), Backend (Go), PostgreSQL database, Redis cache, API-first design, scalable microservices architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Multi Datacenter Model (Section 1)

Hierarchical structure supporting unlimited regions, datacenters, rooms, rows, racks, and devices with strict datacenter isolation. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Inventory System (Section 1)

Tracks full asset lifecycle including procurement, staging, deployment, maintenance, and decommissioning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Rack Management (Section 1)

42U/48U visualization, drag-and-drop placement, utilization tracking, and capacity planning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Cross Connect System (Section 1)

Full lifecycle management of fiber and copper connections including path mapping and dependency tracking. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Visitor Management (Section 1)

QR-based access system with approval workflows, audit logging, and compliance support. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Goods Management (Section 1)

Inbound/outbound logistics tracking using QR with automatic inventory integration. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

QR System Design (Section 1)

Token-based secure QR validation with expiration, backend verification, and audit logging. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

API Design (Section 1)

RESTful API with OpenAPI support enabling integration with external systems. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Security (Section 1)

RBAC, JWT authentication, TLS encryption, audit logs, and compliance features. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

UI/UX Design (Section 1)

Modern dark dashboard, rack visualization, data tables, graph visualization, and mobile QR scanning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Audit & Compliance (Section 1)

Complete audit trail for all operations supporting ISO and enterprise requirements. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Scalability (Section 1)

Horizontal scaling, stateless services, distributed architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Deployment (Section 1)

Docker + Kubernetes ready deployment with CI/CD support. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

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Multi Datacenter Model (Section 14)

Hierarchical structure supporting unlimited regions, datacenters, rooms, rows, racks, and devices with strict datacenter isolation. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Inventory System (Section 14)

Tracks full asset lifecycle including procurement, staging, deployment, maintenance, and decommissioning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Rack Management (Section 14)

42U/48U visualization, drag-and-drop placement, utilization tracking, and capacity planning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Cross Connect System (Section 14)

Full lifecycle management of fiber and copper connections including path mapping and dependency tracking. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Visitor Management (Section 14)

QR-based access system with approval workflows, audit logging, and compliance support. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Goods Management (Section 14)

Inbound/outbound logistics tracking using QR with automatic inventory integration. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

QR System Design (Section 14)

Token-based secure QR validation with expiration, backend verification, and audit logging. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

API Design (Section 14)

RESTful API with OpenAPI support enabling integration with external systems. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Security (Section 14)

RBAC, JWT authentication, TLS encryption, audit logs, and compliance features. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

UI/UX Design (Section 14)

Modern dark dashboard, rack visualization, data tables, graph visualization, and mobile QR scanning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Audit & Compliance (Section 14)

Complete audit trail for all operations supporting ISO and enterprise requirements. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Scalability (Section 14)

Horizontal scaling, stateless services, distributed architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Deployment (Section 14)

Docker + Kubernetes ready deployment with CI/CD support. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Executive Summary (Section 15)

This document defines a full enterprise-grade DCIM platform supporting multi-datacenter operations, QR-based workflows, and operational intelligence. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

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Deployment (Section 15)

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Executive Summary (Section 16)

This document defines a full enterprise-grade DCIM platform supporting multi-datacenter operations, QR-based workflows, and operational intelligence. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Architecture Overview (Section 16)

Frontend (Next.js), Backend (Go), PostgreSQL database, Redis cache, API-first design, scalable microservices architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Multi Datacenter Model (Section 16)

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Inventory System (Section 16)

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Rack Management (Section 16)

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Executive Summary (Section 17)

This document defines a full enterprise-grade DCIM platform supporting multi-datacenter operations, QR-based workflows, and operational intelligence. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Architecture Overview (Section 17)

Frontend (Next.js), Backend (Go), PostgreSQL database, Redis cache, API-first design, scalable microservices architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Multi Datacenter Model (Section 17)

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Inventory System (Section 17)

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Rack Management (Section 17)

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Executive Summary (Section 18)

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Architecture Overview (Section 18)

Frontend (Next.js), Backend (Go), PostgreSQL database, Redis cache, API-first design, scalable microservices architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Multi Datacenter Model (Section 18)

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Inventory System (Section 18)

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Rack Management (Section 18)

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Executive Summary (Section 19)

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Architecture Overview (Section 19)

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Multi Datacenter Model (Section 19)

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Deployment (Section 19)

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Executive Summary (Section 20)

This document defines a full enterprise-grade DCIM platform supporting multi-datacenter operations, QR-based workflows, and operational intelligence. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Architecture Overview (Section 20)

Frontend (Next.js), Backend (Go), PostgreSQL database, Redis cache, API-first design, scalable microservices architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Multi Datacenter Model (Section 20)

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Inventory System (Section 20)

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Rack Management (Section 20)

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Executive Summary (Section 21)

This document defines a full enterprise-grade DCIM platform supporting multi-datacenter operations, QR-based workflows, and operational intelligence. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Architecture Overview (Section 21)

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Executive Summary (Section 22)

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Architecture Overview (Section 22)

Frontend (Next.js), Backend (Go), PostgreSQL database, Redis cache, API-first design, scalable microservices architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Multi Datacenter Model (Section 22)

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Executive Summary (Section 24)

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Executive Summary (Section 25)

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Executive Summary (Section 26)

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Architecture Overview (Section 26)

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Multi Datacenter Model (Section 26)

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Inventory System (Section 93)

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Rack Management (Section 93)

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Hierarchical structure supporting unlimited regions, datacenters, rooms, rows, racks, and devices with strict datacenter isolation. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Inventory System (Section 106)

Tracks full asset lifecycle including procurement, staging, deployment, maintenance, and decommissioning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Rack Management (Section 106)

42U/48U visualization, drag-and-drop placement, utilization tracking, and capacity planning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Cross Connect System (Section 106)

Full lifecycle management of fiber and copper connections including path mapping and dependency tracking. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Visitor Management (Section 106)

QR-based access system with approval workflows, audit logging, and compliance support. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Goods Management (Section 106)

Inbound/outbound logistics tracking using QR with automatic inventory integration. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

QR System Design (Section 106)

Token-based secure QR validation with expiration, backend verification, and audit logging. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

API Design (Section 106)

RESTful API with OpenAPI support enabling integration with external systems. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Security (Section 106)

RBAC, JWT authentication, TLS encryption, audit logs, and compliance features. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

UI/UX Design (Section 106)

Modern dark dashboard, rack visualization, data tables, graph visualization, and mobile QR scanning. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Audit & Compliance (Section 106)

Complete audit trail for all operations supporting ISO and enterprise requirements. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Scalability (Section 106)

Horizontal scaling, stateless services, distributed architecture. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Deployment (Section 106)

Docker + Kubernetes ready deployment with CI/CD support. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Executive Summary (Section 107)

This document defines a full enterprise-grade DCIM platform supporting multi-datacenter operations, QR-based workflows, and operational intelligence. Detailed explanation covering implementation, edge cases, and enterprise scenarios.

Architecture Overview (Section 107)

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Executive Summary (Section 110)

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